9654 - Definitive Constraints on the Star Cluster Environment of Intermediate Mass Black Hole Candidate HLX-1

Cycle: 5, Proposal Category: GO

INVESTIGATORS

Name	*Institution*
Kristen Dage (PI)	**Curtin University**
Dr. Richard M. Plotkin (CoI) (CoPI) (US Admin CoI)	University of Nevada - Reno
Prof. Anil C. Seth (CoI)	University of Utah
Prof. Jay Strader (CoI)	Michigan State University
Prof. Stephen E. Zepf (CoI)	Michigan State University
Dr. Kyle Kremer (CoI)	University of California - San Diego
Prof. Cathryn Trott (CoI)	Curtin University
Anshu Gupta (CoI)	Curtin University
Dr. Behzad Tahmasebzadeh (CoI)	University of Michigan
Dr. Vincent Henault-Brunet (CoI) (CSA Member)	Saint Mary's University
Dr. Kwangmin Oh (CoI)	Michigan State University
Dr. Ana Chies Santos (CoI)	Universidade Federal do Rio Grande do Sul
Prof. Daryl Haggard (CoI) (CSA Member)	McGill University

OBSERVATIONS

Folder	*Observation*	*Label*	*Observing Template*	*Science Target*
Observation Folder				
	1		NIRSpec Fixed Slit Spectroscopy	(1) FWB2009-HLX-1

ABSTRACT

HLX-1 is our best candidate intermediate mass black hole yet. However, the nature of its star cluster environment, and particularly the age, is unclear. HLX-1 may be hosted by a young star cluster, but degeneracy in distinguishing between the effect of the irradiated disk and stellar population age in the SED means that an older population is equally possible. Given the faintness of HLX-1's OIR counterpart (H=24.4), and the large distance to the source, JWST is the only facility that can obtain NIR spectroscopy to probe the age of HLX-1's host cluster. This measurement has important implications for numerical modeling of black holes in star clusters, and will either support the fast growth scenario of black hole formation in dense stellar systems (potentially similar to "little red dots" observed with JWST), or the slow growth scenario (connected to the rate of stellar mass black hole mergers, which has further implications for gravitational wave progenitors). With only a modest investment of time from JWST, we will be able to definitely constrain the age of the host cluster of HLX-1.

OBSERVING DESCRIPTION

We request a two spatial point dithering pattern to correct for hot pixels and to better sample the PSF. We use the strategy of 62 groups per integration and 4 integration per exposure, with one exposure per specification with the NRSIR2RAPID readout pattern. With the requested dithering pattern, this amounts to 15 minutes at each position. Our total time request includes 2.5 hours on target, plus overheads. We propose to use the PRISM/clear filter with the S200A1 slit.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	FWB2009-HLX-1	RA: 01 10 28.3000 (17.6179167d) Dec: -46 04 22.30 (-46.07286d) Equinox: J2000	Epoch of Position: 2000	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.
Category=Stellar Cluster
Description=[Globular star clusters, Young star clusters]

Observation

Proposal 9654, Observation 1

Fri Mar 13 17:03:50 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRSpec Fixed Slit Spectroscopy

Diagnostics

(Observation 1) Warning (Form): Observers are responsible for checking that target acquisition is feasible.

(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
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Acquisition

#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
1	1 FWB2009-HLX-1	WATA	FULL	CLEAR	NRSRAPID	3	1	1	42.947	

Template

HFF Readout Mode	Slit	Subarray
false	S200A1	SUBS200A1

Dithers

#	Primary Dither Positions	Sub-Pixel Pattern
1	2	SPECTRAL

Spectral Elements

#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	PRISM/CLEAR	S200A1	NRS	62	4	1	NONE	6	24	9311.1	